

EXPERIMENT 2

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Experiment 02 – Basic Network Commands and Network Configuration

Aim

To study and execute various Windows networking commands used for configuring, monitoring, and troubleshooting computer networks.

Objectives

- To understand the purpose of common networking commands.
- To verify network connectivity using Ping.
- To examine network configuration using IPConfig.
- To trace packet routes using Tracert.
- To perform DNS lookups using NSLookup.
- To monitor active network connections using Netstat.
- To view ARP cache entries and IP-to-MAC mappings.
- To identify the system hostname.
- To analyze packet loss using PathPing.
- To study routing information and shared network resources using Route and Net Use.

Software Requirements

Software

- Microsoft Windows 11
- Command Prompt (CMD)

Hardware

- Desktop/Laptop

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- Network Interface Card (NIC)
- Internet or LAN Connection

Procedure

1. Open Command Prompt with appropriate permissions.
2. Execute the Ping command and its commonly used variations.
3. Display the system's IP configuration using IPConfig commands.
4. Trace the route to a destination using Tracert.
5. Perform DNS lookups using NSLookup.
6. Display active network connections using Netstat.
7. View the ARP cache using the ARP command.
8. Display the computer name using the Hostname command.
9. Analyze network latency and packet loss using PathPing.
10. View the routing table using the Route command.
11. Display network share information using the Net Use command.
12. Observe and record the outputs of each command.

Commands Executed

Ping

ping google.com

ping -t google.com

ping -n 5 google.com

ping -l 1000 google.com

ping -f google.com

ping -a 8.8.8.8

IPConfig

ipconfig

ipconfig /all

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ipconfig /release

ipconfig /renew

Tracert

tracert google.com

tracert -d google.com

tracert -h 10 google.com

NSLookup

nslookup google.com

nslookup 8.8.8.8

nslookup google.com 8.8.8.8

nslookup -type=A google.com

Netstat

netstat

netstat -a

netstat -n

netstat -o

netstat -ano

ARP

arp -a

arp -d *

Hostname

hostname

PathPing

pathping google.com

Route

route print

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Net Use

net use

Step-wise Output

Step 1 – Ping Command

Step 2 – Ping Variations

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```
PS C:\Users\HP> ping -t google.com

Pinging google.com [2404:6800:4002:81c::200e] with 32 bytes of data:
Reply from 2404:6800:4002:81c::200e: time=48ms
Reply from 2404:6800:4002:81c::200e: time=41ms
Reply from 2404:6800:4002:81c::200e: time=49ms
Reply from 2404:6800:4002:81c::200e: time=70ms
Reply from 2404:6800:4002:81c::200e: time=55ms
Reply from 2404:6800:4002:81c::200e: time=76ms

Ping statistics for 2404:6800:4002:81c::200e:
    Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 41ms, Maximum = 76ms, Average = 56ms
Control-C
PS C:\Users\HP> |
```

```
PS C:\Users\HP> ping -n 5 google.com

Pinging google.com [2404:6800:4002:81c::200e] with 32 bytes of data:
Reply from 2404:6800:4002:81c::200e: time=62ms
Reply from 2404:6800:4002:81c::200e: time=68ms
Reply from 2404:6800:4002:81c::200e: time=46ms
Reply from 2404:6800:4002:81c::200e: time=42ms
Reply from 2404:6800:4002:81c::200e: time=62ms

Ping statistics for 2404:6800:4002:81c::200e:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 68ms, Average = 56ms
PS C:\Users\HP> |
```

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```
PS C:\Users\HP> ping -l 100 google.com

Pinging google.com [2404:6800:4002:81b::200e] with 100 bytes of data:
Reply from 2404:6800:4002:81b::200e: time=78ms
Reply from 2404:6800:4002:81b::200e: time=83ms
Reply from 2404:6800:4002:81b::200e: time=91ms
Reply from 2404:6800:4002:81b::200e: time=86ms

Ping statistics for 2404:6800:4002:81b::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 78ms, Maximum = 91ms, Average = 84ms
PS C:\Users\HP> |
```

```
PS C:\Users\HP> ping -f -l 1472 google.com

Pinging google.com [142.250.182.206] with 1472 bytes of data:
Reply from 10.87.27.170: Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.

Ping statistics for 142.250.182.206:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
PS C:\Users\HP> |
```

Step 3 – IPConfig

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```
C:\Users\Utkar>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4
    Temporary IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05
    Link-local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19
    IPv4 Address. . . . . : 192.168.1.31
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%19
                                192.168.1.1
```

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```
C:\Users\Utkar>ipconfig/all
```

Windows IP Configuration

```
Host Name . . . . . : Utkarsh
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : AC-19-8E-E8-F5-F4
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : AE-19-8E-E8-F5-F3
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Physical Address. . . . . : B2-4A-34-58-0C-C4
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4(Preferred)
Temporary IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05(Preferred)
Link-Local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19(Preferred)
IPv4 Address. . . . . : 192.168.1.31(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 17 July 2026 20:09:07
Lease Expires . . . . . : 18 July 2026 20:11:44
Default Gateway . . . . . : fe80::1%19
                             192.168.1.1
```

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```
C:\Users\Utkar>ipconfig/release
```

```
Windows IP Configuration
```

```
No operation can be performed on Local Area Connection* 1 while it has its media disconnected
```

```
Wireless LAN adapter Local Area Connection* 1:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Local Area Connection* 2:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix . :  
IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4  
Temporary IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05  
Link-local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19  
Default Gateway . . . . . : fe80::1%19
```

```
C:\Users\Utkar>ipconfig /renew
```

```
Windows IP Configuration
```

```
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.  
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
```

```
Wireless LAN adapter Local Area Connection* 1:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Local Area Connection* 2:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix . :  
IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4  
Temporary IPv6 Address. . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05  
Link-local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19  
IPv4 Address. . . . . : 192.168.1.53  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::1%19  
192.168.1.1
```

Step 4 – Tracert

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```
C:\Users\Utkar>tracert python-quiz.pages.dev
```

```
Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2d23]  
over a maximum of 30 hops:
```

1	1 ms	1 ms	1 ms	2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
2	*	*	*	Request timed out.
3	3 ms	3 ms	2 ms	2404:a800:1a00:804::cd
4	13 ms	13 ms	14 ms	2404:a800:88::134
5	15 ms	15 ms	*	2404:a800:0:29::cf
6	13 ms	13 ms	17 ms	2606:4700:310c::ac42:2d23

```
Trace complete.
```

```
C:\Users\Utkar>tracert -d python-quiz.pages.dev
```

```
Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2d23]  
over a maximum of 30 hops:
```

1	1 ms	1 ms	1 ms	2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
2	*	*	*	Request timed out.
3	2 ms	2 ms	2 ms	2404:a800:1a00:804::cd
4	13 ms	13 ms	13 ms	2404:a800:88::134
5	*	15 ms	15 ms	2404:a800:0:29::cf
6	13 ms	11 ms	11 ms	2606:4700:310c::ac42:2d23

```
Trace complete.
```

```
C:\Users\Utkar>tracert -h 5 python-quiz.pages.dev
```

```
Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2edd]  
over a maximum of 5 hops:
```

1	1 ms	1 ms	1 ms	2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
2	*	*	*	Request timed out.
3	3 ms	2 ms	2 ms	2404:a800:1a00:804::cd
4	11 ms	13 ms	11 ms	2404:a800:88::134
5	12 ms	13 ms	12 ms	2404:a800:0:29::cf

```
Trace complete.
```

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```
C:\Users\Utkar>tracert -?
```

```
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
           [-R] [-S srcaddr] [-4] [-6] target_name
```

Options:

-d	Do not resolve addresses to hostnames.
-h maximum_hops	Maximum number of hops to search for target.
-j host-list	Loose source route along host-list (IPv4-only).
-w timeout	Wait timeout milliseconds for each reply.
-R	Trace round-trip path (IPv6-only).
-S srcaddr	Source address to use (IPv6-only).
-4	Force using IPv4.
-6	Force using IPv6.

Step 5 – NSLookup

```
C:\Users\Utkar>nslookup python-quiz.pages.dev
```

```
Server: 192.168.1.1
```

```
Address: 192.168.1.1
```

```
Non-authoritative answer:
```

```
Name: python-quiz.pages.dev
```

```
Addresses: 2606:4700:310c::ac42:2edd
           2606:4700:310c::ac42:2d23
           172.66.46.221
           172.66.45.35
```

```
C:\Users\Utkar>nslookup -type=A python-quiz.pages.dev
```

```
Server: 192.168.1.1
```

```
Address: 192.168.1.1
```

```
Non-authoritative answer:
```

```
Name: python-quiz.pages.dev
```

```
Addresses: 172.66.45.35
           172.66.46.221
```

Step 6 – Netstat

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```
C:\Users\Utkar>netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49367	Utkarsh:64463	ESTABLISHED
TCP	127.0.0.1:49696	Utkarsh:49697	ESTABLISHED
TCP	127.0.0.1:49697	Utkarsh:49696	ESTABLISHED
TCP	127.0.0.1:49699	Utkarsh:49700	ESTABLISHED
TCP	127.0.0.1:49700	Utkarsh:49699	ESTABLISHED
TCP	127.0.0.1:49707	Utkarsh:49708	ESTABLISHED
TCP	127.0.0.1:49708	Utkarsh:49707	ESTABLISHED
TCP	127.0.0.1:56058	Utkarsh:56059	ESTABLISHED
TCP	127.0.0.1:56059	Utkarsh:56058	ESTABLISHED
TCP	127.0.0.1:63101	Utkarsh:63102	ESTABLISHED
TCP	127.0.0.1:63102	Utkarsh:63101	ESTABLISHED
TCP	127.0.0.1:64076	Utkarsh:64464	ESTABLISHED
TCP	127.0.0.1:64463	Utkarsh:49367	ESTABLISHED
TCP	127.0.0.1:64464	Utkarsh:64076	ESTABLISHED
TCP	192.168.1.53:50647	192:8009	ESTABLISHED
TCP	192.168.1.53:50648	ec2-107-20-21-3:https	ESTABLISHED
TCP	192.168.1.53:50649	192:8009	ESTABLISHED
TCP	192.168.1.53:52688	ec2-52-45-140-188:https	ESTABLISHED
TCP	192.168.1.53:54189	192:8009	ESTABLISHED
TCP	192.168.1.53:56093	192:8009	ESTABLISHED
TCP	192.168.1.53:56094	192:8009	ESTABLISHED
TCP	192.168.1.53:56098	192:8009	ESTABLISHED
TCP	192.168.1.53:56522	192:8009	ESTABLISHED
TCP	192.168.1.53:56600	ec2-18-97-36-65:https	ESTABLISHED
TCP	192.168.1.53:57783	192:8008	ESTABLISHED
TCP	192.168.1.53:57784	192:8008	ESTABLISHED
TCP	192.168.1.53:57789	192:8008	ESTABLISHED
TCP	192.168.1.53:57790	192:8008	ESTABLISHED
TCP	192.168.1.53:60739	lb-140-82-113-26-iad:https	ESTABLISHED
TCP	192.168.1.53:60741	ec2-98-85-208-136:https	ESTABLISHED
TCP	192.168.1.53:60951	192:8009	ESTABLISHED
TCP	192.168.1.53:61178	ec2-18-97-36-44:https	ESTABLISHED

EXPERIMENT 2

```
C:\Users\Utkar>netstat -o
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	127.0.0.1:49367	Utkarsh:64463	ESTABLISHED	11860
TCP	127.0.0.1:49696	Utkarsh:49697	ESTABLISHED	1912
TCP	127.0.0.1:49697	Utkarsh:49696	ESTABLISHED	1912
TCP	127.0.0.1:49699	Utkarsh:49700	ESTABLISHED	3228
TCP	127.0.0.1:49700	Utkarsh:49699	ESTABLISHED	3228
TCP	127.0.0.1:49707	Utkarsh:49708	ESTABLISHED	5300
TCP	127.0.0.1:49708	Utkarsh:49707	ESTABLISHED	5300
TCP	127.0.0.1:56058	Utkarsh:56059	ESTABLISHED	1372
TCP	127.0.0.1:56059	Utkarsh:56058	ESTABLISHED	1372
TCP	127.0.0.1:63101	Utkarsh:63102	ESTABLISHED	26820
TCP	127.0.0.1:63102	Utkarsh:63101	ESTABLISHED	26820
TCP	127.0.0.1:64076	Utkarsh:64464	ESTABLISHED	11860
TCP	127.0.0.1:64463	Utkarsh:49367	ESTABLISHED	12244
TCP	127.0.0.1:64464	Utkarsh:64076	ESTABLISHED	12244
TCP	192.168.1.53:50647	192:8009	ESTABLISHED	19924
TCP	192.168.1.53:50648	ec2-107-20-21-3:https	ESTABLISHED	7172
TCP	192.168.1.53:50649	192:8009	ESTABLISHED	12244
TCP	192.168.1.53:52688	ec2-52-45-140-188:https	ESTABLISHED	7172
TCP	192.168.1.53:54189	192:8009	ESTABLISHED	32420
TCP	192.168.1.53:56093	192:8009	ESTABLISHED	16008
TCP	192.168.1.53:56094	192:8009	ESTABLISHED	16008
TCP	192.168.1.53:56098	192:8009	ESTABLISHED	12244
TCP	192.168.1.53:56522	192:8009	ESTABLISHED	32420
TCP	192.168.1.53:56600	ec2-18-97-36-65:https	ESTABLISHED	32420
TCP	192.168.1.53:57783	192:8008	ESTABLISHED	16008
TCP	192.168.1.53:57784	192:8008	ESTABLISHED	16008
TCP	192.168.1.53:57789	192:8008	ESTABLISHED	12244
TCP	192.168.1.53:57790	192:8008	ESTABLISHED	12244
TCP	192.168.1.53:60739	lb-140-82-113-26-iad:https	ESTABLISHED	32420
TCP	192.168.1.53:60741	ec2-98-85-208-136:https	ESTABLISHED	30016
TCP	192.168.1.53:60951	192:8009	ESTABLISHED	19924
TCP	192.168.1.53:61178	ec2-18-97-36-44:https	ESTABLISHED	32420

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```
C:\Users\Utkar>netstat -r
```

Interface List

```
6...ac 19 8e e8 f5 f4 .....Microsoft Wi-Fi Direct Virtual Adapter
10...ae 19 8e e8 f5 f3 .....Microsoft Wi-Fi Direct Virtual Adapter #2
19...b2 4a 34 58 0c c4 .....Intel(R) Wi-Fi 6E AX211 160MHz
1.....Software Loopback Interface 1
```

IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.53	35
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
192.168.1.0	255.255.255.0	On-link	192.168.1.53	291
192.168.1.53	255.255.255.255	On-link	192.168.1.53	291
192.168.1.255	255.255.255.255	On-link	192.168.1.53	291
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.1.53	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.1.53	291

Persistent Routes:

None

IPv6 Route Table

Active Routes:

If	Metric	Network Destination	Gateway
19	291	::/0	fe80::1
1	331	::1/128	On-link
19	291	2401:4900:1c6e:c2fe::/64	On-link
19	291	2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05/128	On-link
19	291	2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4/128	On-link
19	291	fe80::/64	On-link
19	291	fe80::948e:7197:14dc:d497/128	On-link
1	331	ff00::/8	On-link
19	291	ff00::/8	On-link

Persistent Routes:

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```
C:\Users\Utkar>netstat -e
Interface Statistics
```

	Received	Sent
Bytes	79068230	346668168
Unicast packets	2160390	1681470
Non-unicast packets	86334	10134
Discards	0	0
Errors	0	0
Unknown protocols	0	0

Step 7 – ARP

```
C:\Users\Utkar>arp -a
```

```
Interface: 192.168.1.53 --- 0x13
Internet Address      Physical Address      Type
192.168.1.1           b0-8b-92-94-f9-f2    dynamic
192.168.1.7           3c-c5-dd-6b-c6-37    dynamic
192.168.1.11          78-b3-9f-e4-2e-95    dynamic
192.168.1.33          30-68-93-28-04-18    dynamic
192.168.1.49          50-bb-b5-f2-43-e0    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

Step 8 – Hostname

```
C:\Users\Utkar>hostname
Utkarsh
```

Step 9–

PathPing

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```
C:\Users\Utkar>pathping python-quiz.pages.dev

Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2edd]
over a maximum of 30 hops:
  0  Utkarsh [2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05]
  1  2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
  2  * * *
Computing statistics for 25 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
  0                               100/ 100 =100%   Utkarsh [2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05]
  1  ---      100/ 100 =100%   0/ 100 = 0%     2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2

Trace complete.
```

Step 10 – Route Print

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```
C:\Users\Utkar>route print
=====
Interface List
 6...ac 19 8e e8 f5 f4 .....Microsoft Wi-Fi Direct Virtual Adapter
10...ae 19 8e e8 f5 f3 .....Microsoft Wi-Fi Direct Virtual Adapter #2
19...b2 4a 34 58 0c c4 .....Intel(R) Wi-Fi 6E AX211 160MHz
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.1.1      192.168.1.53     35
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.1.0                 255.255.255.0    On-link          192.168.1.53     291
192.168.1.53                255.255.255.255  On-link          192.168.1.53     291
192.168.1.255               255.255.255.255  On-link          192.168.1.53     291
224.0.0.0                   240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                   240.0.0.0        On-link          192.168.1.53     291
255.255.255.255             255.255.255.255  On-link          127.0.0.1        331
255.255.255.255             255.255.255.255  On-link          192.168.1.53     291
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
19      291 ::/0                      fe80::1
1       331 ::1/128                  On-link
19      291 2401:4900:1c6e:c2fe::/64 On-link
19      291 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05/128
                                           On-link
19      291 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4/128
                                           On-link
19      291 fe80::/64                  On-link
19      291 fe80::948e:7197:14dc:d497/128
                                           On-link
1       331 ff00::/8                  On-link
19      291 ff00::/8                  On-link
=====
Persistent Routes:
None
```

Step 11 – Net Use

EXPERIMENT 2

```
C:\Users\Utkar>  
  
C:\Users\Utkar>net use  
New connections will be remembered.  
  
There are no entries in the list.
```

Learning Outcome

After completing this experiment, I was able to:

- Understand the purpose and functionality of essential Windows networking commands.
- Test network connectivity and identify communication issues using Ping.
- Examine and manage TCP/IP configuration using IPConfig.
- Trace packet routes and identify intermediate network devices using Tracert.
- Resolve domain names and verify DNS services using NSLookup.
- Monitor network connections and protocol statistics using Netstat.
- Analyze ARP cache entries and understand IP-to-MAC address mapping.
- Identify the system hostname and network identity.
- Detect packet loss and network delays using PathPing.
- View routing information and shared network resources using Route and Net Use.
- Develop practical troubleshooting **skills for diagnosing common network problems.**